

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

Course Code: CSE 219 Course Title: Signals and Linear Systems

Dear everyone, this assignment is primarily intended for anyone who may have missed a class test. However, any of you in this course may also submit the assignment. We will consider the best two scores among the two class tests and this assignment. Please submit a scanned PDF of your solution to this assignment on Moodle. Thank you!

Submission Deadline: 14 August, 2026**Marks: 20**

1. **Convolution.** Consider the following two continuous-time signals: [6]

$$x(t) = u(t - 1) - u(t - 4)$$

and

$$y(t) = u(t - 2) - u(t - 3),$$

where $u(t)$ denotes the continuous-time unit-step function.

Please compute the convolution output

$$z(t) = x(t) * y(t).$$

Finally, draw the output signal $z(t)$ clearly and label the important values on the t -axis and the vertical axis wherever necessary.

2. **Fourier Series.** Please consider the continuous-time periodic signal $x(t)$ with period $T = 2$ as [6]

$$x(t) = \begin{cases} t^2, & 0 \leq t < 1, \\ 0, & 1 \leq t < 2, \end{cases} \quad x(t + T) = x(t).$$

Please find the complex exponential Fourier series coefficients a_k , where

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}.$$

3. **Fourier Transform.** Please first prove that [6]

$$e^{-t}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{1 + j\omega}.$$

Then, using this result and the Fourier transform properties given below, find the Fourier transform of

$$te^{-|t|}.$$

You may use the following Fourier transform properties. If

$$x(t) \xleftrightarrow{\mathcal{F}} X(\omega),$$

then

$$x(-t) \xleftrightarrow{\mathcal{F}} X(-\omega) \quad (\text{time reversal})$$

$$\alpha x_1(t) + \beta x_2(t) \xleftrightarrow{\mathcal{F}} \alpha X_1(\omega) + \beta X_2(\omega) \quad (\text{linearity})$$

and

$$tx(t) \xleftrightarrow{\mathcal{F}} j \frac{dX(\omega)}{d\omega} \quad (\text{multiplication by } t).$$

Please clearly show how you use these properties.

4. Thank you once again for your attention and dedication! May you always remain honest, kind, caring, divine, and innocent. Please take care of your health, drink adequate water, and please always have sound sleep and adequate rest. Wishing you all the best! [2]